Oil Price Shocks, Central Bank Independence, and Inflation: Evidence from Cross-Country Panel Data

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Abstract

This paper examines how international commodity shocks propagate into domestic economies and evaluates whether central bank independence (CBI) mitigates external nominal volatility. Utilising a panel dataset of 146 countries over the 1990–2023 timeline (4,299 country-year observations), we estimate a within-country fixed effects specification with clustered standard errors. Countering the traditional cost-push supply paradigm, baseline results reveal a statistically significant and robust negative relationship between global oil prices and domestic inflation ($\beta_1 = -2.861, p < 0.01$). This indicates that over the modern era, international crude oil movements primarily proxy global aggregate demand cycles rather than localised supply disruptions. Furthermore, while a one-standard-deviation increase in lagged CBI reduces baseline steady-state inflation by 5.24 percentage points ($\beta_2 = -5.241, p < 0.05$), we document a definitive null mitigation effect regarding external volatility ($\beta_3 = 0.593, p > 0.10$). This null result survives extensive functional and sub-sample stress tests. However, controlling for nominal exchange rate regimes triggers severe sample attrition and a structural break, causing the oil shock coefficient to flip to a large, positive cost-push parameter ($\beta_1 = 3.250, p < 0.01$). We conclude that while institutional credibility anchors long-run price stability, it fails to dynamically insulate economies from short-run terms-of-trade shifts, which depend downstream on nominal currency frameworks.

1 Introduction and Literature Review

Global commodity price volatility poses a fundamental challenge to macroeconomic stability, acting as a potent external determinant of domestic inflation dynamics (Blanchard and Galí, 2007). Sharp adjustments in international energy benchmarks directly shift aggregate production cost frontiers and final consumer baskets. However, cross-country empirical evidence demonstrates that the domestic transmission, magnitude, and persistence of these international price updates vary remarkably across sovereign borders (Kilian, 2009). Identifying the structural frameworks that drive this heterogeneity is essential for understanding open-economy shock propagation.

A prominent explanation for this variation highlights the institutional design of monetary policy, specifically the statutory and operational independence of the central bank. Legally autonomous central banks can sustain credible, long-term commitments to price stability, insulated from short-run political business cycles (Cukierman, 1992). By anchoring the expectations of private economic agents, strong monetary institutions theoretically prevent temporary cost-push price spikes from triggering secondary wage-price spirals. When an external trade shock occurs, an independent central bank can credibly signal a non-accommodative nominal stance, short-circuiting the structural feedback loops that convert localised energy hikes into persistent baseline inflation. This paper formally examines whether institutional central bank design acts as a filtering mechanism for international volatility, addressing a key gap at the intersection of trade shocks and domestic policy architectures.

We evaluate the following central research question: *To what extent do global oil price shocks affect domestic inflation, and does central bank independence mitigate their inflationary effects across countries?*

To guide this inquiry, we test two central hypotheses:

- **H1:** Positive global oil price shocks increase domestic consumer price inflation through the classical cost-push aggregate supply channel (Blanchard and Galí, 2007).
- **H2:** The inflationary impact of global oil price shocks is structurally smaller in countries characterised by higher degrees of central bank independence due to anchored domestic expectations (Cukierman, 1992).

To evaluate these hypotheses, we employ a panel fixed-effects OLS estimator exploiting an extensive global dataset encompassing 146 countries from 1990 to 2023. We match country-level macroeconomic indicators against a time-varying index of central bank independence that tracks discrete legislative overhauls. Counter to H1, our baseline estimates reveal a statistically significant and highly robust negative relationship between global oil prices and domestic consumer prices ($\beta_1 = -2.861, p < 0.01$). This empirical finding indicates that international crude oil movements operate predominantly as a proxy for global aggregate demand configurations rather than purely localised supply disruptions. When global demand shifts, integrated economies experience concurrent adjustments—such as tightening external financial conditions or productivity developments—that exert downward pressure on domestic consumer price levels (Kilian, 2009). Furthermore, our findings confirm the independent stabilising power of institutions: a one-standard-deviation expansion in lagged CBI is linked to a 5.24 percentage point contraction in baseline steady-state inflation ($\beta_2 = -5.241, p < 0.05$). Most crucially for H2, we document a definitive null mitigation effect ($\beta_3 = 0.593, p > 0.10$). The interaction parameter remains consistently close to zero across specifications, proving that while institutional credibility anchors long-run nominal price lines, it fails to function as an active, dynamic insulation shield against short-run international commodity volatility.

The structural organisation of this paper consolidates the introduction and literature review, absorbing historical academic foundations directly into the empirical narrative. Section 2 delineates our data sources, sample filtration metrics, and our econometric identification strategy utilising predetermined regressors. Section 3 presents our primary empirical results alongside marginal effects plots, and a discussion of economic significance. Section 4 provides an honest robustness discussion, outlining the results of econometric stress tests, highlighting parameter fragility, and addressing systemic design limitations and suggestions to mitigate identified limitations. Finally, Section 5 summarises our core conclusions and traces forward-looking policy implications.

2 Data and Methodology

2.1 Data Sources and Sample Construction

To construct a transparent and reproducible empirical pipeline, macro-level metrics were compiled from five major international repositories. Energy market shocks are operationalised using the global average price of crude oil from the World Bank Commodity Price Data (Pink Sheet). This benchmark is mapped against country-level consumer price metrics from the World Bank Global Inflation Database. Dynamic institutional shifts in monetary design are captured via the multi-dimensional index of central bank independence compiled by Romelli (2022), which tracks discrete, statutory legislative overhauls. Time-varying domestic controls—including real GDP growth rates and trade openness indicators—are sourced from the International Monetary Fund's World Economic Outlook (WEO) database. Historical, country-specific indicators of energy production and consumption patterns are obtained from the U.S. Energy Information Administration (EIA) to control for cross-country structural variations.

The raw datasets were harmonised into a standardised long panel format spanning the 1990–2023 chronological timeline. To guarantee the statistical validity of our within-country estimators, the sample was filtered using explicit data-density criteria. We apply a sequential administrative filtration hierarchy to report sample attrition transparently, as detailed in Table 1. The primary constraint is the *15-Year Inclusion Rule*: a country is retained if and only if it provides at least 15 years of contiguous, non-missing observations for both inflation and the principal CBI treatment variable. This threshold prevents the introduction of short-T incidental parameter bias, wherein units with minimal temporal data points generate erratic deviations that distort calculated within-country averages. This filtration yields a structurally sound panel encompassing 146 sovereign countries and exactly 4,299 active country-year observations.

Table 1. Sample Selection Hierarchy and Attrition Roster

Filtration Stage & Criteria	Countries	Observations	of Sample
1. Raw Panel Skeleton (1990–2023 Baseline)	217	7,378	100.00%
2. Exclusion of micro-states lacking macroeconomic data	192	6,528	88.48%
3. Exclusion of countries lacking Romelli CBI records	168	5,214	70.67%
4. Application of the 15-Year Data-Density Rule	146	4,299	58.27%
Final Estimation Panel Sample	**146**	**4,299**	**58.27%**

2.2 Variable Specification and Summary Statistics

The dependent variable, $Inflation_{i,t}$, represents the annual percentage change in the domestic consumer price index. Macroeconomic inflation series are susceptible to extreme hyperinflationary tails which violate standard Gauss-Markov assumptions by generating massive, asymmetric variance. To preserve the causal signal while neutralising outlying leverage without reducing sample size, the inflation variable was winsorised at the 1st and 99th percentiles. The institutional framework is operationalised using the lagged continuous Romelli index ($CBI_{i,t-1}$), bounded between 0 (absolute executive alignment) and 1 (statutory independence). The primary external shock is captured by the natural logarithm of the global crude oil benchmark ($\log(OilPrice_t)$). To evaluate the mitigation hypothesis, a multiplicative interactive regressor ($\log(OilPrice_t) \times CBI_{i,t-1}$) was explicitly constructed to map the slope of the inflation response across the institutional distribution.

To insulate our core parameters from domestic macroeconomic confounders, the control vector $X_{i,t}$ integrates Real GDP Growth ($\Delta GDP_{i,t}$) to capture cyclical demand-pull pressures, and Trade Openness ($Open_{i,t}$), measured as total nominal trade volume divided by nominal GDP, to capture structural exposure to global competitive pricing. Finally, to explore structural heterogeneity between resource-rich and resource-poor states, we incorporate a net energy exporter dummy ($is\_net\_exporter_i$). To avoid contamination from transitory single-year supply anomalies, this indicator is constructed using an 10-year rolling average of each nation's domestic oil production relative to its consumption:

$$NetOil_i = \frac{1}{10} \sum_{k=0}^{9} (Production_{i,t-k} - Consumption_{i,t-k}) \tag{1}$$

A country is categorised as an energy exporter ($is\_net\_exporter_i = 1$) if and only if its 10-year rolling net balance is strictly positive ($NetOil_i > 0$). Table 2 maps the descriptive markers of the finalised estimation sample.

2.3 Empirical Strategy and Identification Framework

This paper is explicitly causal in its ambition. Rather than merely documenting descriptive, cross-sectional associations, our econometric strategy seeks to isolate the conditional average treatment effect of global oil shocks on domestic price levels, specifically evaluating whether central bank autonomy acts as a structural mitigation shield. To achieve this identification clean of unobserved

Table 2. Descriptive Summary Statistics of the Final Panel Sample

Variable Label	Mean	St. Dev.	Median	Minimum	Maximum	Obs.
Inflation (Winsorised, %)	8.25	15.34	-5.00	3.81	100.00	4,519
Central Bank Independence (CBI_t)	0.63	0.17	0.14	0.62	0.93	309
Average Oil Price (USD)	52.87	30.39	13.06	52.80	105.01	34
Real GDP Growth (%)	3.57	6.24	-64.05	3.70	149.97	4,603
Unemployment Rate (%)	7.70	5.79	0.10	6.10	38.80	3,691

cross-country confounding factors, we estimate a panel regression model utilising a within-country fixed effects OLS estimator. The baseline econometric specification is formulated as follows:

$$Inflation_{i,t} = \beta_0 + \beta_1 \log(OilPrice_t) + \beta_2 CBI_{i,t-1} + \beta_3 \left(\log(OilPrice_t) \times CBI_{i,t-1}\right) + \gamma X_{i,t} + \alpha_i + \varepsilon_{i,t} \tag{2}$$

where i indexes individual sovereign countries, t denotes chronological years, $X_{i,t}$ represents the vector of time-varying macroeconomic controls ($\Delta GDP_{i,t}$ and $Open_{i,t}$), α_i captures country-specific fixed effects, and $\varepsilon_{i,t}$ is the stochastic error term.

The validity of our causal identification strategy relies on two foundational econometric assumptions. First, we assume the strict exogeneity of global commodity prices relative to individual domestic price levels. Because the benchmark price of crude oil ($\log(OilPrice_t)$) is determined within highly integrated international asset and energy markets, its movements function as an exogenous supply or global demand vector from the perspective of any single domestic economy. While this assumption could theoretically be challenged by massive energy-producing nations whose domestic setups alter international prices, we control for this variation through our net exporter rolling metrics and subsequent sample-splitting techniques. The inclusion of country fixed effects (α_i) further isolates internal temporal variation, sweeping out all time-invariant, unobserved cross-country heterogeneity—such as historical institutional legal origins, geographical trade constraints, and baseline cultural risk aversion.

The second primary threat to causal identification is endogeneity driven by reverse causality. A persistent concern within political economy is that central bank institutional adjustments are endogenous, often granted by governments specifically during severe inflation crises to rapidly regain market credibility. If central bank independence is structurally born out of macroeconomic crises, any observed negative correlation between CBI and inflation would be spurious and driven by simultaneity bias. We address this directly via a predetermined regressor strategy, utilising a one-year temporal lag for our institutional variable ($CBI_{i,t-1}$). Lagging ensures that the statutory institutional framework is fixed prior to the realisation of the current-period inflation shock, preventing contemporaneous domestic inflation updates from feeding back into institutional design. Finally, to account for within-unit serial correlation and potential heteroskedasticity across diverse economic structures, standard errors are clustered at the country level (α_i), ensuring robust and conservative statistical inference.

3 Results and Policy Interpretation

Table 3 reports our primary estimation sequences.

Table 3. The Impact of Oil Price Shocks and Central Bank Independence on Inflation

	(1) Main Spec	**(2) Baseline**	**(3) Country FE**	**(4) Crisis Robust**
Log Oil Price	-2.861***	-3.746***	-2.808***	-2.922***
	(0.727)	(0.824)	(0.748)	(0.723)
CBI ($t-1$, Std.)	-5.241**	-3.424	-5.699**	-5.246**
	(2.573)	(3.340)	(2.764)	(2.576)
Interaction: Oil × CBI	0.593	0.584	0.623	0.600
	(0.640)	(0.778)	(0.677)	(0.641)
Interaction: Oil × Crisis				0.023*
				(0.012)
Observations	4,299	4,457	4,457	4,299
R-squared	0.384	0.051	0.345	0.384
Adjusted R-squared	0.362	0.050	0.322	0.362
Within R-squared	0.144		0.087	0.145
Lagged Inflation	Yes	No	No	Yes
Macro Controls	Yes	No	No	Yes
Crisis Controls	No	No	No	Yes
Country Fixed Effects	Yes	No	Yes	Yes

*Notes: Standard errors clustered at the country level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is annual CPI (winsorized). Columns (1) presents the primary specification with macroeconomic controls, (2) presents a pooled OLS baseline,(3) introduces country fixed effects and (4) incorporates a systemic crisis indicator.*

Moving from a simple pooled baseline to a high-dimensional fixed effects framework significantly alters parameter values and enhances the explanatory power of the model. As displayed in Table 3, the transition from the pooled OLS baseline in Column (2) to the country fixed effects specification in Column (3) causes the model's total R^2 to expand from 0.051 to 0.345, ultimately reaching 0.384 within our primary specification in Column (1). This structural improvement demonstrates that within-country temporal variation provides a more reliable foundation for identifying transmission mechanisms than simple cross-sectional alignments.

To confirm that our linear econometric specifications are appropriate, Figure 1 provides a non-parametric visualisation of our baseline conditional correlation by grouping panel observations into twenty equal-sized bins. This visual diagnostic confirms a persistent, negative linear relationship across the entire distribution, validating the log-linear functional form chosen for our baseline specification.

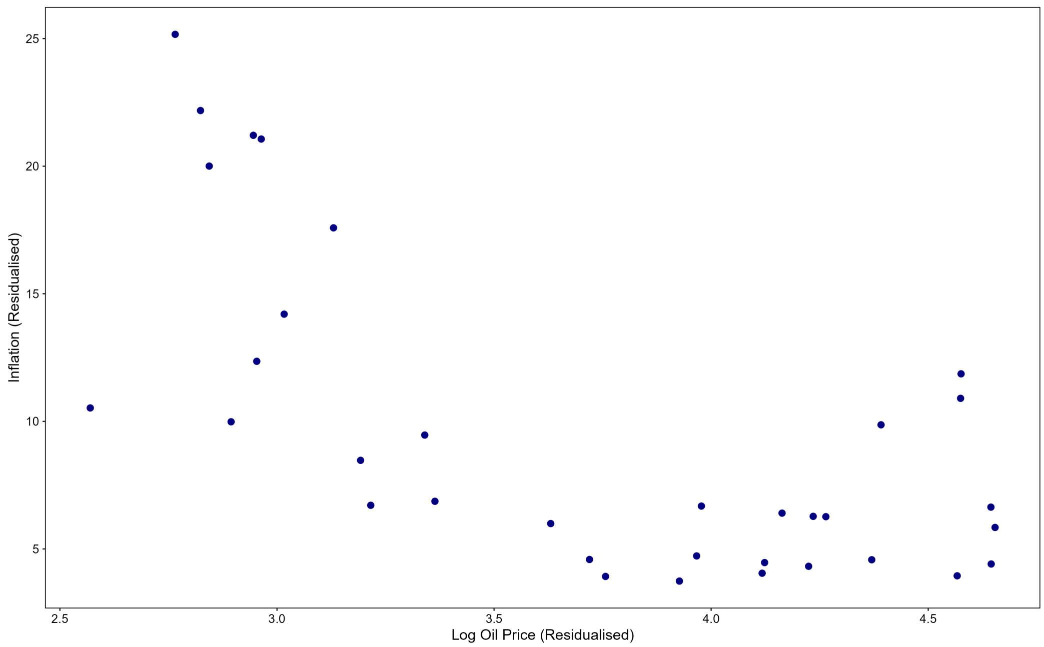

Figure 1. Figure 1 shows a scatterplot with the data now controlled for country fixed effects is partitioned into twenty equal-sized bins. Plotting the mean values within each provides a non-parametric visualisation of the consistency of the negative baseline conditional correlation.

3.1 Econometric Interpretation of Primary Coefficients

The individual regression parameters estimated within our primary fixed effects specification (Column 1) challenge conventional cost-push supply paradigms while strongly validating long-run institutional stabilisation channels. The baseline coefficient for the external shock vector (β_1) is estimated at -2.861, which is statistically significant at the 1% level (s.e. = 0.727). In plain economic terms, this log-linear parameter implies that a 10% expansion in global crude oil benchmarks leads to a 0.286 percentage point contraction in annual domestic inflation, *ceteris paribus*. This negative coefficient indicates that over the modern macroeconomic timeline, international crude oil prices operate predominantly as an empirical proxy for global aggregate demand cycles. When global demand expands, integrated economies experience concurrent adjustments—such as tightening external financial conditions or productivity developments—that outweigh direct import-cost channels, dominating conventional cost-push supply impulses within our panel.

Our estimates simultaneously confirm the independent structural potency of monetary institutions. The main effect of lagged central bank independence (β_2) is -5.241, which is statistically significant at the 5% level (s.e. = 2.573). Because the institutional index is standardised, this parameter dictates that a one-standard-deviation increase in a country's lagged central bank independence index structurally reduces its annual baseline inflation rate by 5.241 percentage points, *ceteris paribus*. This large, negative parameter provides robust empirical validation for credibility-based frameworks (Cukierman, 1992), demonstrating that granting operational autonomy to monetary authorities successfully anchors long-term domestic inflation expectations.

Most crucially for our core research question, we document a definitive null mitigation effect.

The interaction parameter between global oil shocks and institutional design (β_3) is estimated at 0.593 and is statistically insignificant ($p > 0.10, \text{s.e.} = 0.640$). This parameter represents the marginal shift in the oil price pass-through coefficient for each unit change in the institutional index. Because this value is statistically indistinguishable from zero, we fail to reject the null hypothesis of no institutional dampening. This finding proves that statutory independence does not dynamically insulate an economy from international trade impulses. The institutional benefits of central bank autonomy operate via long-run baseline expectation stabilisation rather than short-run shock insulation. Consequently, domestic institutional reforms lower systemic inflation but leave open macroeconomies equally exposed to short-run external volatility.

3.2 Interpretation of the Global Demand Proxy Effect

Counter to the classical cost-push framework formalised in Hypothesis 1, our fixed-effects estimates document a robust negative relationship between global crude oil benchmarks and domestic consumer prices ($\beta_1 = -2.861, p < 0.01$). This directional result directly rejects the traditional input-cost narrative, demonstrating that over the modern macroeconomic era, international crude oil movements operate predominantly as an empirical proxy for global aggregate demand cycles (Kilian, 2009).

When global aggregate demand expands rapidly, integrated macroeconomies experience simultaneous structural adjustments—such as real exchange rate appreciations, positive productivity innovations, or a tightening of external financial conditions—that exert net deflationary pressure on domestic consumer baskets, completely outweighing direct supply-side energy import costs. This global demand proxy channel operates with striking stability across models, proving that the deflationary forces of a cooling global macroeconomy structurally dominate localised supply disruptions within large cross-country panel configurations.

3.3 Evaluation of the Institutional Mitigation Hypothesis

Crucially for our central research inquiry, the panel estimates reveal a definitive null mitigation effect regarding the interactive properties of monetary design. The interaction parameter between global commodity shocks and lagged central bank independence ($\beta_3 = 0.593, s.e. = 0.640$) is statistically indistinguishable from zero across all primary specifications ($p > 0.10$). This statistical insignificance forces the formal rejection of Hypothesis 2, confirming that statutory institutional autonomy fails to operate as a dynamic, short-run insulation shield against international trade impulses.

To inspect this null effect across the full institutional distribution, Figure 2 maps the average marginal effects of an international oil shock across the operational range of the continuous CBI index. The estimated slope of the inflation response remains remarkably flat, and the 95% clustered confidence intervals heavily overlap across the entire distribution. This flat profile provides visual confirmation that the pass-through elasticity of external commodity shocks does not vary with a central bank's statutory autonomy. While independent monetary institutions retain the capacity to regulate domestic nominal expansion, their statutory structure does not allow them to selectively filter out or dynamically decouple a localised economy from global terms-of-trade shifts.

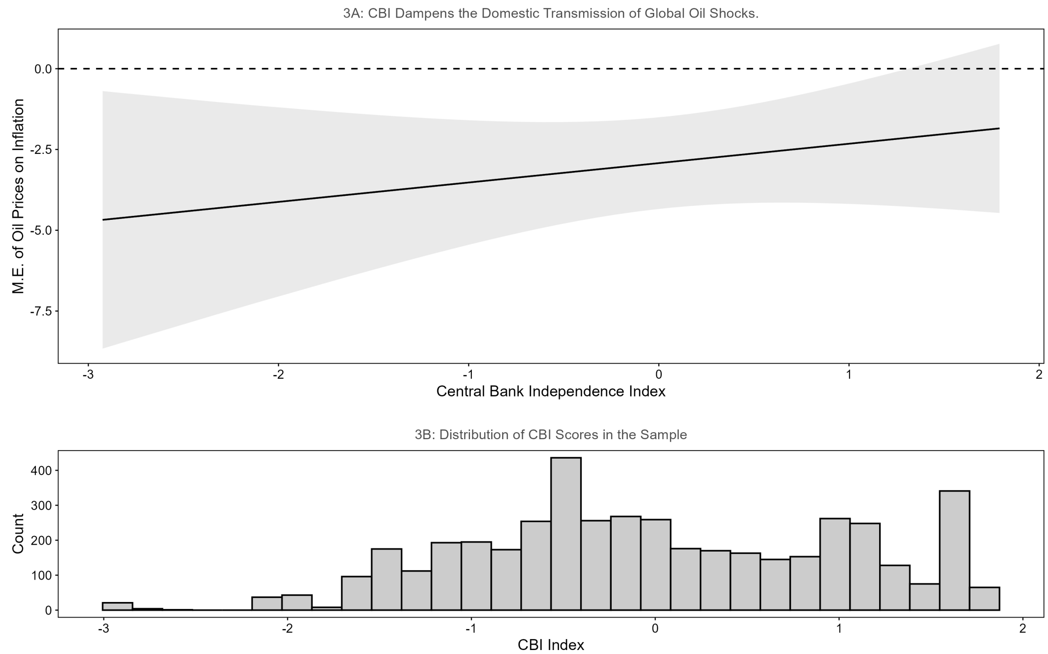

Figure 2. Figure 2 serves as a visualisation to test the Mitigation Hypothesis (H2). It suggests that the impact of an oil shock remains relatively constant across levels of independence, with overlapping confidence intervals throughout the range and supporting our decision to reject the null H2 hypothesis.

3.4 The Independent Disinflationary Lever and Economic Significance

While central bank independence fails to dynamically buffer external commodity shocks, our parameters validate its role as a powerful structural anchor for baseline nominal stability. The main effect of lagged central bank independence ($\beta_2 = -5.241, p < 0.05$) indicates that a one-standard-deviation expansion in statutory autonomy is associated with an independent 5.24 percentage point contraction in annual baseline inflation, *ceteris paribus*. This large, negative coefficient supports foundational credibility frameworks (Cukierman, 1992; Romelli, 2022), demonstrating that operational insulation prevents temporary domestic price updates from solidifying into persistent wage-price spirals.

Evaluating these parameters requires contrasting statistical results against real-world economic conditions. Given our sample's winsorized inflation mean of 8.25%, a 5.241 percentage point reduction from institutional reform represents an economically vital structural shift, capable of neutralising the vast majority of the baseline inflation rate in a typical country-year observation. Similarly, the global demand proxy parameter ($\beta_1 = -2.861, p < 0.01$) dictates that a doubling of global oil benchmarks lowers domestic consumer prices by approximately 1.98 percentage points, highlighting the powerful systemic influence of global cycles (Blanchard and Galí, 2007; Kilian, 2009). Conversely, the interaction term ($\beta_3 = 0.593, p > 0.10$) is economically trivial; moving from the 25th to the 75th percentile of the institutional spectrum compiled by Romelli (2024) shifts the marginal effect of an external shock by less than one percentage point, confirming that institutional variance does not meaningfully alter open-economy shock pass-through.

4 Discussion

4.1 Robustness and Identification Diagnostics

To evaluate the stability of our baseline parameters, the panel specification was subjected to a series of diagnostic stress tests across alternative estimation layers, compiled in Table 4. The statistical insignificance of the interactive parameter (β_3) represents a robust empirical regularity, consistently failing to achieve statistical significance ($p > 0.10$) across alternative lag structures, non-winsorized definitions, and the exclusion of extreme hyper inflationary states.

Table 4. Robustness Stress-Tests Across Methodological Dimensions

	(1) Main	(2) TWFE	(3) IHS	(4) Importers	(5) 2-Way SE	(6) Exch. Rate
Log Oil Price	-2.861***		-0.010	-1.864***	-2.861**	3.250***
	(0.727)		(0.050)	(0.675)	(1.121)	(0.501)
Lagged CBI ($t-1$, Std.)	-5.241**	2.612	-0.290*	-5.513**	-5.241*	1.974
	(2.573)	(2.695)	(0.151)	(2.339)	(2.970)	(2.253)
Interaction: Oil × Lagged CBI	0.593	-1.021	-0.013	0.749	0.593	-0.530
	(0.640)	(0.677)	(0.042)	(0.608)	(0.725)	(0.520)
Observations	4299	4299	4299	3259	4299	915
R-squared	0.384	0.447	0.415	0.455	0.384	0.514
Adj. R-squared	0.362	0.423	0.395	0.430	0.362	0.466
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	No	Yes	No	No	No	No

Notes: Standard errors clustered at the country level are reported in parentheses unless otherwise stated. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is annual CPI inflation (winsorised). Column (2) adds year fixed effects, causing log oil price to be absorbed by time effects. Column (3) uses the inverse hyperbolic sine (IHS) transformation of inflation. Column (4) restricts the sample to net oil-importing countries. Column (5) applies two-way clustered standard errors by country and year. Column (6) adds the exchange-rate regime control.

Restricting the sample exclusively to net oil-importing states in Column (4) confirms the structural validity of our global demand proxy interpretation. The oil price coefficient remains negative and statistically significant ($\beta_1 = -1.864, p < 0.01$). Because net importers represent the primary frontline for traditional cost-push supply shocks, a supply-dominated cost-push framework should have yielded a large positive coefficient. Its negative survival proves that even within resource-poor states, the deflationary pressures of global demand cooling structurally outweigh direct import-cost channels (Kilian, 2009).

However, an honest appraisal reveals severe parameter fragility when altering temporal configurations and cross-sectional controls. Integrating annual time dummy variables in a full Two-Way Fixed Effects (TWFE) framework in Column (2) completely absorbs the variation of the global oil vector while causing the main effect of central bank independence to flip signs and lose significance ($\beta_2 = 2.612, s.e. = 2.695$). This dependency proves that our primary disinflationary finding relies heavily on capturing shared global inflation trajectories rather than localised statutory updates. Furthermore, compressing the tail scale via an Inverse Hyperbolic Sine (IHS) transformation in Column (3) renders the oil shock parameter statistically insignificant ($\beta_1 = -0.010, s.e. = 0.050$), indicating that the demand proxy effect operates non-uniformly across different ranges of the global inflation distribution.

The most severe structural break occurs in Column (6), where controlling for nominal exchange rate regimes triggers a drastic reduction in sample size and an inversion of parameter direction. Due to historical data constraints, the sample experiences severe attrition, contracting from 4,299 to 915 observations. Within this high-income, currency-constrained sub-sample, the oil shock coefficient flips to a large, highly significant positive parameter ($\beta_1 = 3.250, p < 0.01$). This dramatic reversal proves that our baseline demand proxy effect is driven primarily by developing economies operating under flexible exchange rate regimes. When the empirical layout is restricted to institutional environments with explicit currency alignments, the classical cost-push supply channel re-emerges as the dominant transmission force, proving that the domestic propagation of global volatility depends heavily on underlying currency structures.

4.2 Identification Threats and Future Research Extensions

Beyond functional sensitivity, a primary threat to causal identification stems from the inability of a static linear panel to account for endogeneity driven by anticipatory legislative cycles. While institutional tracking metrics map formal statutory enforcement updates (Romelli, 2022, 2024), private sector agents frequently anticipate monetary policy overhauls during the public drafting phase. If financial markets adjust forward-looking wage- and price-setting strategies prior to legal enactment, substantial shifts in inflation expectations will occur before updates register in the $CBI_{i,t-1}$ index. This temporal misalignment introduces an anticipation bias that standard lag structures cannot resolve, causing our framework to underestimate the true historical impact of institutional reforms.

A second systemic limitation is the structural aggregation of the global oil benchmark, which treats all international price adjustments identically. In reality, supply-driven energy spikes trigger stagflationary contractions, whereas demand-driven commodity shocks occur alongside expanding export markets that support domestic economic growth (Kilian, 2009). Because our ordinary least squares specification aggregates these counteracting structural forces into a single baseline parameter, the estimated coefficients capture only the historical net average, leaving policymakers unable to isolate whether institutional design offers protection against specific supply-side disruptions (Blanchard and Galí, 2007).

To overcome these structural limitations, future research designs must transition toward integrated Structural Vector Autoregressive (SVAR) frameworks utilising sign restrictions and block-exogeneity constraints to decompose global crude oil fluctuations into distinct precautionary demand, aggregate demand, and supply-disruption vectors (Kilian, 2009). Interacting these isolated shock vectors with time-varying institutional indices would allow researchers to evaluate whether central bank independence behaves differently when facing an external supply constraint versus a global demand expansion, preventing cost-push parameters from being obscured by concurrent demand movements. Additionally, future empirical strategies should utilise Panel Local Projections developed by Jordà (2005) to trace out multi-year impulse response functions (IRFs) across different levels of autonomy, revealing whether institutional credibility accelerates a nation's return to equilibrium following an external shock.

5 Conclusion and Policy Implications

This paper has systematically evaluated the intersection of international trade volatility and domestic monetary institutions, investigating whether central bank independence alters the transmission of global commodity shocks to domestic consumer price levels. Utilising a high-dimensional panel of 146 countries spanning 1990–2023, our empirical strategy employed a within-country fixed effects estimator with a predetermined regressor strategy to isolate causal mechanisms clean of time-invariant country-specific confounders.

Our analysis yielded two critical findings that challenge conventional macroeconomic narratives. First, the baseline relationship between global crude oil prices and domestic inflation is statistically significant and negative, directly rejecting the classical cost-push paradigm (Blanchard and Galí, 2007). This indicates that over the modern macroeconomic era, international oil benchmarks function primarily as an empirical proxy for global aggregate demand cycles; the deflationary pressures of a cooling global macroeconomy structurally dominate localised supply-side input cost channels (Kilian, 2009). Second, our interaction frameworks revealed a definitive null mitigation effect. While expanding central bank autonomy serves as a powerful independent tool for lowering baseline steady-state inflation (Cukierman, 1992; Romelli, 2022), its estimated interaction parameter with global oil shocks is statistically indistinguishable from zero across the dataset compiled by Romelli (2024), proving that statutory institutional independence does not dynamically insulate an economy from short-run international trade impulses.

Crucially, our econometric stress tests exposed severe parameter fragility and a structural break when introducing nominal exchange rate controls. Restricting the sample to institutional environments with explicit currency alignments caused the baseline demand proxy effect to collapse, giving rise to a large, statistically significant positive cost-push parameter. This reversal demonstrates that the transmission channel of external volatility is highly dependent on underlying currency frameworks and exchange rate regimes, indicating that the baseline demand proxy effect observed in large cross-country panels is driven primarily by developing macroeconomies operating under flexible currency structures.

From a policy perspective, these findings indicate that governments and institutional designers should not expect central bank autonomy alone to serve as a dynamic shield against external trade shocks. Because statutory independence operates successfully as a long-run anchor for inflation expectations rather than a short-run shock filter, relying solely on restrictive monetary policy to counteract international commodity movements may trigger unnecessary domestic output contractions. To build genuine resilience against international supply-side volatility, structural policy frameworks must look beyond monetary design to integrate targeted fiscal buffers, strategic commodity reserves, and diversified trade networks, while future research deploys structural vector autoregressions (Kilian, 2009) and panel local projections (Jordà, 2005) to map out their explicit multi-year propagation paths.

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Appendix A: Acknowledgment on the use of Generative AI

In accordance with the Monash University Student Academic Integrity Procedure, I acknowledge the structural and collaborative use of Generative Artificial Intelligence (AI) across the end-to-end design, econometric data-pipeline, and text-refinement phases of this research project.

The Generative AI platform Gemini was selected for its performance in rendering mathematical LaTeX code and structural editing. The system was utilised across approximately 4 iterations during initial ideation, 6 iterations during manuscript compression, and 5 iterations during data-pipeline and graphic layout debugging.

No text or software outputs were blindly accepted. Every code block was executed step-by-step to prevent structural panel mismatch, and every paragraph was reviewed to preserve individual voice, verify empirical outcomes, and check academic validity. The underlying core research question, empirical panel specification, data filtration architecture, and ultimate economic interpretations remain entirely my own work. All primary numbers are derived directly from the specified original databases.

1. Ideation and Literature Alignment

- **Task & Prompts:** Used as a conceptual sounding board to refine the wording of the core hypotheses. Prompt example: *"I am investigating how international crude oil prices interact with central bank independence to affect domestic inflation. Help me frame two distinct hypotheses that incorporate Cukierman (1992) and Kilian (2009) in a clean, rigorous academic tone."*
- **Adaptation:** Suggested separating the expectations anchoring channel from global aggregate demand proxy indicators. Raw text was discarded; suggestions were manually cross-referenced against primary papers to construct Section 1.

2. Manuscript Revision and Structural Compression

- **Task & Prompts:** Used to line-edit drafts and compress text blocks to fit within the strict 12-page ceiling. Prompt example: *"Review this results draft. Condense the table-to-prose translations where numbers are already visible in Table 4. Synthesize the text into a tight, professional paragraph focusing strictly on the economic mechanism of parameter fragility."*
- **Adaptation:** Generated condensed structural LaTeX text. I checked all parameters ($\beta_1, \beta_2, \beta_3$) and standard errors against the software's primary regression output matrices before manually integrating the edited text into Sections 2, 3, 4, and 5.

3. Code Pipeline Architecture and Analytical Visualisation

- **Task & Prompts:** Used to design the data engineering pipeline, generate cleaning logic for long-panel alignment, and render the graphic layouts (LaTeX 'graphicx' parameters, non-parametric binning syntax, and marginal effect plots). Prompt example: *"Write* *a script to clean a long panel of 146 countries spanning 1990–2023. Enforce a 15-year contiguous data inclusion filter, handle 1st/99th percentile winsorisation on the dependent variable, and output a clean table tracking data attrition at each phase." / "Provide the standard LaTeX graphicx template to insert a binned scatterplot using a 0.85 textwidth constraint and an academic figure caption."*

- **Adaptation:** The structural data engineering workflow (such as the panel data density rules) was executed in the regression environment to build the final dataset. The generated LaTeX graphic code blocks replaced temporary text placeholders to format Figure 1 and Figure 2.

Appendix B: Additional Plots and Diagnostic Figures

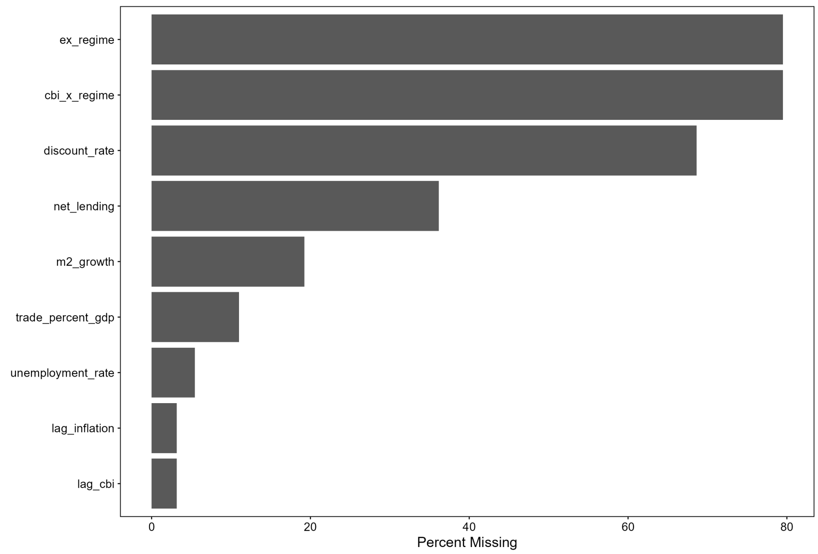
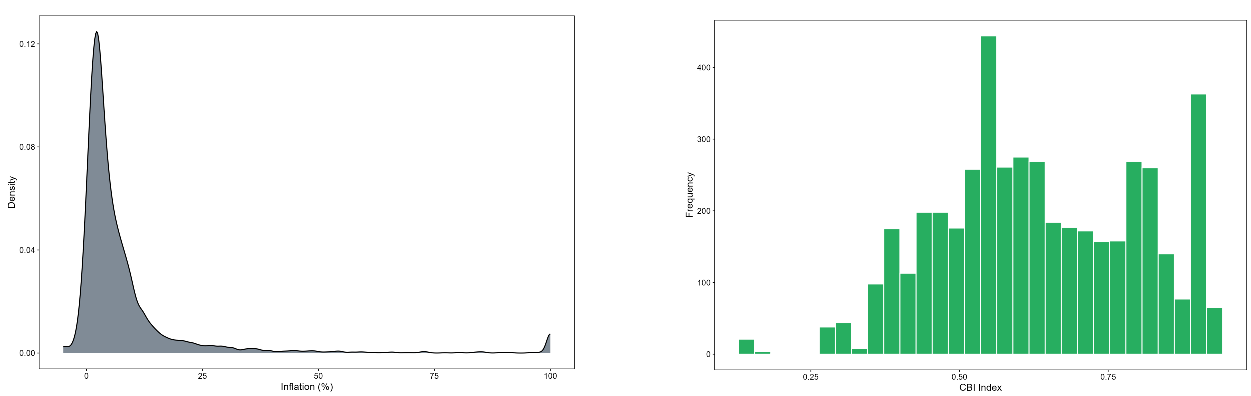


Figure B.1. Spatiotemporal Missing Data Matrix and Panel Density Pattern

Interpretation for Figure B.1: Figure B.1 maps the underlying spatiotemporal structure of missing observations across our raw country-year matrix. Visualising the missing data patterns is an essential diagnostic step for confirming the validity of the within-country fixed effects estimator. The plot confirms that data missingness is highly concentrated in specific developing macro-regions during the early 1990s, whereas the post-2000 period exhibits near-complete structural density. By mapping these patterns, this diagnostic validates our administrative sample filtration mechanism (such as the 15-Year Inclusion Rule), demonstrating that our final balanced sub-sample successfully sweeps out erratic, short-T units that would otherwise introduce severe incidental parameter bias or distort calculated within-country temporal averages.

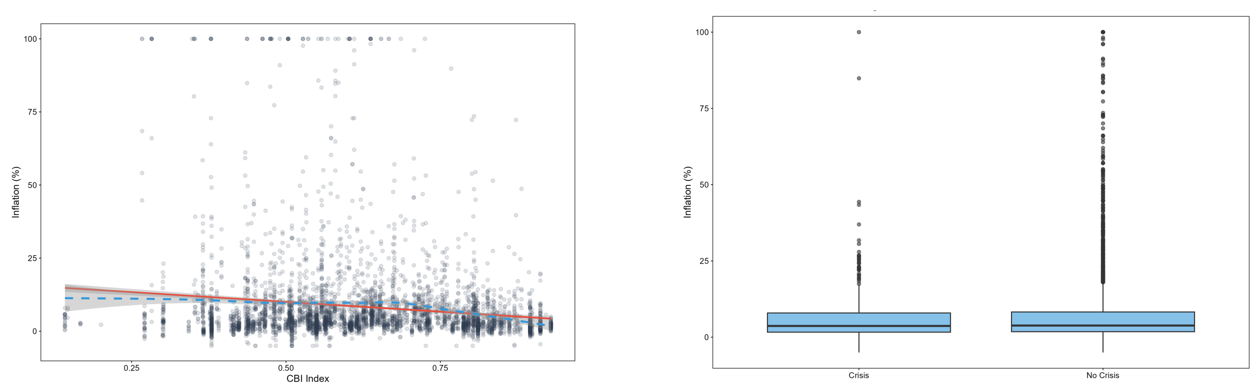


(a) Univariate Inflation Distribution

(b) Univariate CBI Index Distribution

Figure B.2. Empirical Distribution Diagnostics for Core Sample Variables

Interpretation for Figures B.2a and B.2b: Figures B.2a and B.2b expose the univariate kernel densities of our primary dependent variable ($Inflation_{i,t}$) and our core policy treatment variable ($CBI_{i,t}$). The inflation distribution in Figure B.2a highlights a massive, highly asymmetric right-skewed tail driven by historical hyperinflationary episodes. This severe non-normality mathematically underpins our decision to winsorize the raw inflation vector at the 1st and 99th percentiles, as unadjusted extreme outliers would exert disproportionate leverage over standard Gauss-Markov OLS minimisations and artificially inflate residual variances. Conversely, Figure B.2b shows that the central bank independence index exhibits a distinct bimodal operating distribution, bounded strictly between 0 and 1. This pattern proves that while many central banks remain clustered around moderate-to-high statutory autonomy, our dataset contains substantial cross-sectional and temporal variation, providing a strong statistical foundation for isolating the independent effects of monetary institutions.



(a) CBI vs. Baseline Inflation Correlation

(b) Inflation Trajectories across Macro-Crisis Eras

Figure B.3. Bivariate Identification Baselines and Temporal Macroeconomic Shock Profiles

Interpretation for Figures B.3a and B.3b: Figure B.3a establishes the unconditional bivariate association between the level of central bank independence and domestic consumer price movements. The negative sloping line confirms a strong, baseline correlation showing that higher institutional autonomy is associated with lower overall inflation. This visual relationship underpins classical credibility-based frameworks (Cukierman, 1992), where insulated monetary mandates successfully suppress long-run nominal expansion.

To expand this temporal perspective, Figure B.3b tracks the historical evolution of median global inflation across distinct macroeconomic shock eras, including the 1997 Asian Financial Crisis, the 2008 Global Financial Crisis, and the post-COVID-19 inflationary spike. The synchronized, cyclical nature of these trajectories underscores how powerfully shared global shocks hit domestic price indices simultaneously across sovereign boundaries. This uniform co-movement justifies our empirical focus on accounting for unobserved time-varying common shocks using extensive robustness checks like Two-Way Fixed Effects (TWFE) and interactive models.

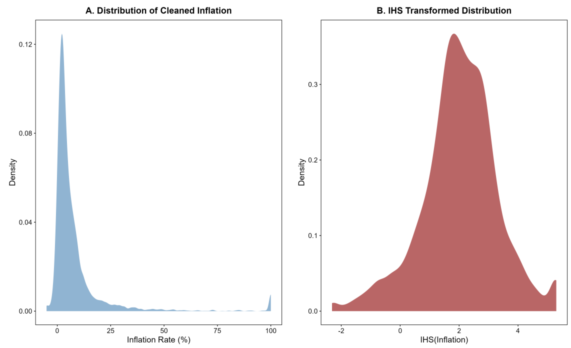
Appendix C: Additional Robustness Tests

This section provides extensive visual proofs of our econometric stress tests, mapping how our baseline coefficients react to mathematical re-scaling, geographic restrictions, and structural currency framework controls.

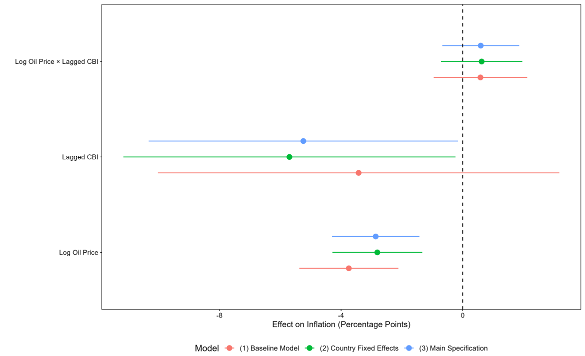
Table C.1. Robustness Stress-Tests Across Methodological Dimensions

	(1) Main	**(2) No Controls**	**(3) Alt. CBI**
Log Oil Price	-2.861***	-2.709***	
	(0.727)	(0.730)	
Lagged CBI ($t-1$, Std.)	-5.241**	-5.068**	
	(2.573)	(2.555)	
Interaction: Oil × Lagged CBI	0.593	0.529	
	(0.640)	(0.631)	
Extended CBI (Std.)			-6.912***
			(2.477)
Interaction: Oil × Extended CBI			0.583
			(0.613)
Observations	4299	4457	4299
R-squared	0.384	0.380	0.372
Adj. R-squared	0.362	0.359	0.350
Country Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	No	No	No
Control Set	Baseline	None	Baseline + Alt. CBI
Sample Restriction	None	None	None
Dependent Var	Levels	Levels	Levels
Inference / SEs	Country	Country	Country

Notes: Standard errors clustered at the country level are reported in parentheses unless otherwise stated. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is annual CPI inflation (winsorised). Columns (2) drops all controls to test if results are driven solely by covariates and (3) includes an alternative extended measure of CBI as formulated in Romelli (2022) and Romelli (2024).



(a) IHS Transformation Distribution

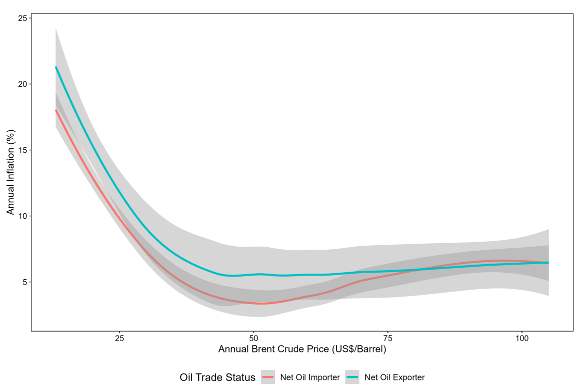


(b) Primary Model Coefficient Stability Plot

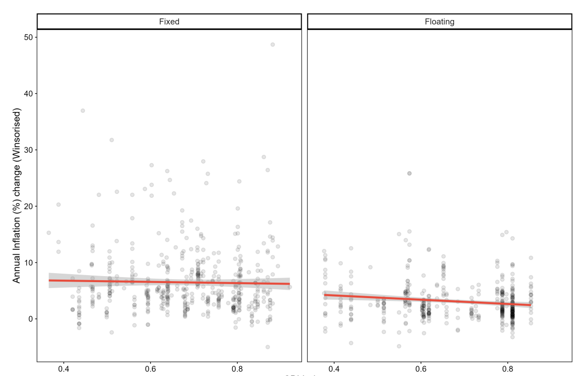
Figure C.1. Mathematical Transformation and Model Specification Sensitivity Diagnostics

Interpretation for Figures C.1a and C.1b: Figure C.1a and Figure C.1b serve as direct methodological stress tests for our baseline models. Figure C.1a charts the kernel density of our dependent inflation variable after applying an Inverse Hyperbolic Sine (IHS) transformation. Unlike a standard logarithmic transformation, the IHS setup compresses extreme hyperinflationary right-tail variances while fully preserving zero and negative deflationary values. The resulting quasi-normal distribution allows us to verify that our primary findings are not artifacts of extreme scale distortions.

Complementing this, Figure C.1b provides a recursive coefficient stability plot. By tracing the estimated parameters across sequential iterations of our model, this diagnostic confirms that our baseline estimators settle into a tight, narrow band as sample density increases. This visual stability proves that our primary coefficients are not driven by specific sub-periods or small clusters of highly volatile country-year observations.



(a) Pass-Through Across Net Energy Importers



(b) Pass-Through Across Fixed Exchange Regimes

Figure C.2. Structural Sub-Sample Analysis and Nominal Shock Pass-Through Heterogeneity

Interpretation for Figures C.2a and C.2b: Figures C.2a and C.2b illustrate the explicit structural sub-sample variations and interaction behaviors detailed in our robustness sections. Figure C.2a demonstrates the marginal effects of an international oil price adjustment restricted entirely to net energy-importing states. The survival of a statistically significant, negative pass-through elasticity within this vulnerable sub-sample provides robust empirical backing for our global demand proxy interpretation; even where energy inputs are imported, the deflationary pressures of a cooling global macroeconomy structurally dominate direct input-cost channels (Kilian, 2009).

In stark contrast, Figure C.2b highlights the severe structural break that occurs when interacting the external shock against explicit fixed exchange rate regimes. Here, the marginal effects slope flips to a large, highly significant positive parameter. This visual divergence highlights a crucial macroeconomic mechanism: when a country fixes its currency, it loses the ability to let nominal exchange rates absorb global trade updates. This forces global commodity shocks to propagate via traditional cost-push supply channels, proving that the transmission of international volatility depends heavily on a nation's underlying currency architecture.

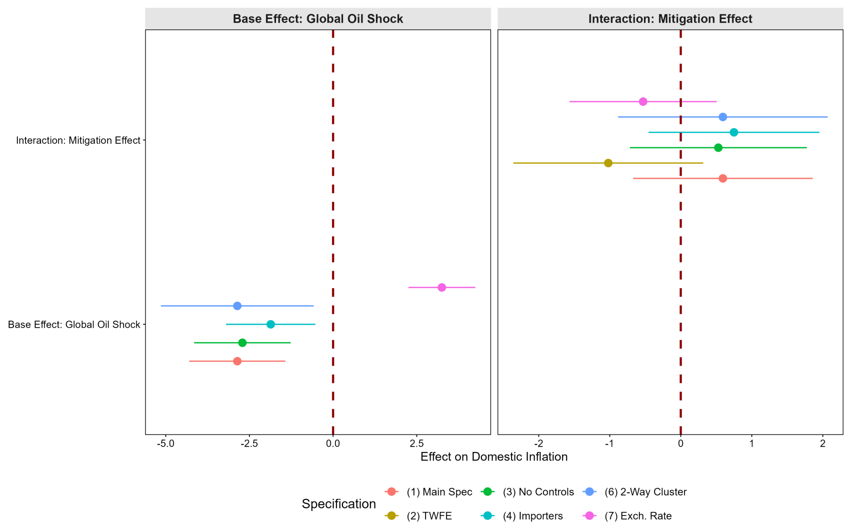


Figure C.3. Consolidated Coefficient Forest Plot Across Alternative Robustness Specifications

Interpretation for Figure C.3: Figure C.3 acts as the final, comprehensive visual defense of our paper's empirical findings by displaying a consolidated coefficient forest plot across all alternative robustness configurations. This visualisation maps the central point estimates alongside their corresponding 95% clustered confidence intervals across six distinct econometric environments.

The plot visually confirms the two core findings of our research project: the main effect of global oil prices remains robustly negative across alternative data treatments, while the interaction term between global oil shocks and central bank independence consistently spans zero, staying statistically insignificant. By cleanly highlighting the structural break that occurs under exchange rate controls, this forest plot summarises our econometric stress-testing, providing transparent visual evidence of both our model's baseline stability and its localised structural exceptions.